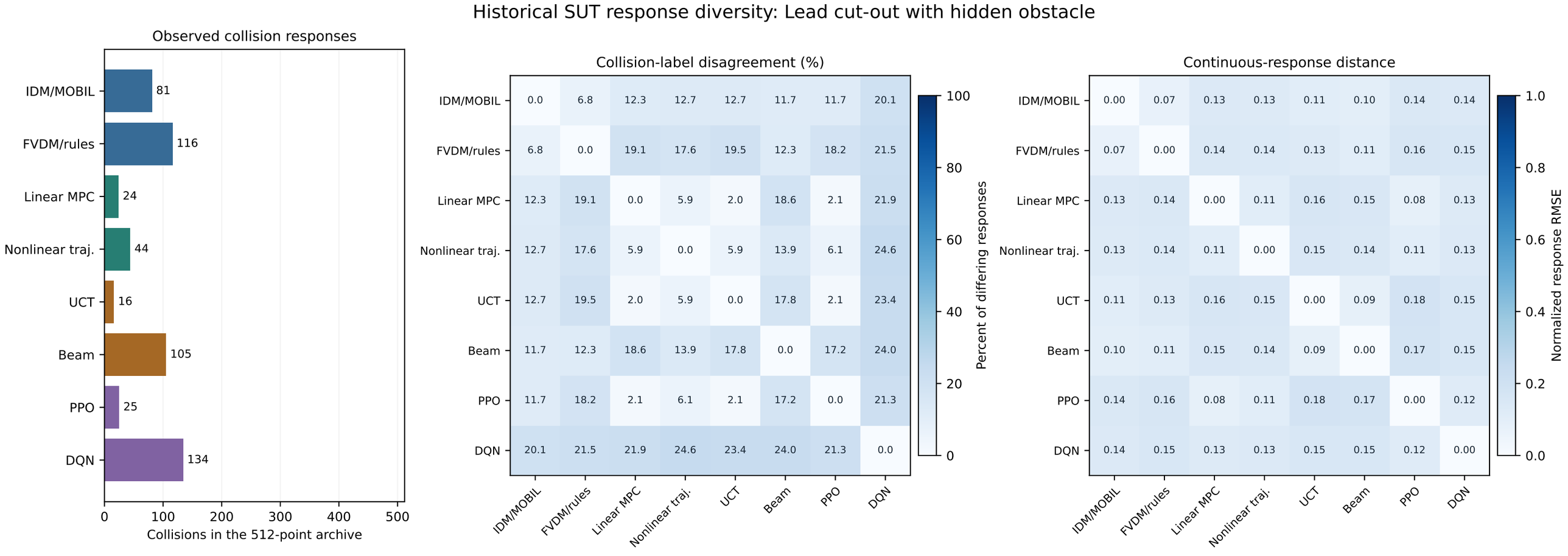
Historical SUT response diversity: Lead cut-out with hidden obstacle
Observed collision responses
IDM/MOBIL
FVDM/rules
Linear MPC
Nonlinear traj.
UCT
Beam
PPO
DQN
81
116
24
44
16
105
25
134
0
100
200
300
400
500
Collisions in the 512-point archive
Collision-label disagreement (%)
0.0 6.8 12.3 12.7 12.7 11.7 11.7 20.1
6.8 0.0 19.1 17.6 19.5 12.3 18.2 21.5
12.3 19.1 0.0 5.9 2.0 18.6 2.1 21.9
12.7 17.6 5.9 0.0 5.9 13.9 6.1 24.6
12.7 19.5 2.0 5.9 0.0 17.8 2.1 23.4
11.7 12.3 18.6 13.9 17.8 0.0 17.2 24.0
11.7 18.2 2.1 6.1 2.1 17.2 0.0 21.3
20.1 21.5 21.9 24.6 23.4 24.0 21.3 0.0
Percent of differing responses
0
20
40
60
80
100
Continuous-response distance
0.00 0.07 0.13 0.13 0.11 0.10 0.14 0.14
0.07 0.00 0.14 0.14 0.13 0.11 0.16 0.15
0.13 0.14 0.00 0.11 0.16 0.15 0.08 0.13
0.13 0.14 0.11 0.00 0.15 0.14 0.11 0.13
0.11 0.13 0.16 0.15 0.00 0.09 0.18 0.15
0.10 0.11 0.15 0.14 0.09 0.00 0.17 0.15
0.14 0.16 0.08 0.11 0.18 0.17 0.00 0.12
0.14 0.15 0.13 0.13 0.15 0.15 0.12 0.00
Normalized response RMSE
0.0
0.2
0.4
0.6
0.8
1.0